

MALMSTROM AFB, MT, USA

WMO: 727755

Lat: 47.517N

Lon: 111.183W

Elev: 1058

StdP: 89.24

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24112

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-25.7	-22.7	-28.8	0.3	-24.9	-25.7	0.4	-22.6	15.2	6.7	14.0	6.1	2.6	240	0.665

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.1	34.6	16.9	32.6	16.2	30.7	15.6	18.6	30.2	17.4	28.8	16.5	27.9	4.5	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.7	11.9	22.1	13.3	10.8	20.8	12.2	10.1	20.0	56.7	30.1	52.9	29.0	49.9	27.6	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
13.7	12.1	10.9	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-31.0	37.4	3.9	2.1	-33.8	38.9	-36.1	40.2	-38.3	41.4	-41.2	42.9
			-30.9	20.4	3.8	2.0	-33.7	21.8	-35.9	23.0	-38.0	24.1	-40.8	25.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.1	-2.7	-3.3	2.4	6.9	11.8	16.7	21.5	20.5	15.5	8.1	1.9	-2.6
	DBStd	10.85	9.08	9.43	7.48	5.06	4.47	3.82	3.61	3.78	4.89	5.63	7.87	8.54
	HDD10.0	1918	396	375	243	116	31	2	0	0	11	99	252	393
	HDD18.3	3979	652	607	493	342	206	74	12	23	111	317	493	650
	CDD10.0	1237	2	2	9	24	88	203	358	325	175	41	9	1
	CDD18.3	256	0	0	0	1	4	26	111	89	25	1	0	0
	CDH23.3	3773	0	0	1	19	102	404	1555	1244	423	24	0	0
	CDH26.7	1642	0	0	0	1	22	132	749	572	162	4	0	0
(14) Wind	WSAvg	4.7	5.8	5.1	5.2	4.7	4.1	4.1	3.7	3.7	4.0	4.9	5.8	5.7
(15) Precipitation	PrecAvg	329	10	9	15	34	55	70	34	32	28	21	12	10
	PrecMax	538	27	30	35	61	110	135	148	67	50	56	30	31
	PrecMin	221	1	0	1	8	8	28	2	5	2	2	0	1
	PrecStd	73	6	7	10	16	22	32	33	19	13	12	8	8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.2	17.1	21.3	26.0	29.5	33.4	38.1	36.5	33.9	26.4	20.6	15.3
		MCWB	6.8	7.5	9.6	12.2	15.0	17.4	20.6	17.3	15.9	13.2	10.0	6.3
	2%	DB	12.8	13.8	17.8	22.2	26.2	30.3	35.2	34.1	30.9	22.7	16.5	12.2
		MCWB	5.3	5.7	7.7	11.0	13.9	16.6	17.5	16.3	14.6	11.4	7.8	5.2
	5%	DB	10.4	11.2	15.3	18.9	23.6	27.6	33.0	32.2	28.3	19.4	14.2	9.9
		MCWB	3.9	4.5	6.5	9.0	12.8	15.4	16.7	15.6	14.0	10.1	6.7	3.7
	10%	DB	8.5	8.8	12.8	16.0	21.1	25.4	31.0	30.1	25.4	16.8	12.2	7.7
		MCWB	2.9	3.3	5.3	7.5	11.6	14.5	16.0	15.1	13.0	8.8	5.7	2.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.8	8.3	10.1	13.1	16.3	19.7	21.8	18.6	17.3	13.9	10.6	7.4
		MCDB	14.8	15.5	19.9	23.0	28.5	28.7	36.2	32.6	29.4	24.3	19.2	14.8
	2%	WB	5.5	6.3	8.4	11.1	14.6	17.8	19.2	17.4	15.8	12.1	8.7	5.5
		MCDB	12.1	12.8	16.8	21.2	24.1	27.4	31.7	30.5	28.0	21.0	15.7	11.3
	5%	WB	4.3	4.8	7.0	9.6	13.4	16.6	17.9	16.5	14.7	10.6	7.2	4.1
		MCDB	9.9	10.8	14.6	18.0	22.0	25.5	29.4	29.3	26.3	18.2	13.7	9.2
	10%	WB	3.0	3.4	5.6	7.9	12.3	15.3	16.8	15.7	13.5	9.2	5.8	2.7
		MCDB	7.9	8.7	12.3	15.2	19.8	23.7	28.2	27.7	24.3	16.0	11.7	7.4
(35) Mean Daily Temperature Range	5% DB	MDBR	10.6	11.0	12.0	12.2	12.5	13.6	16.1	15.9	14.3	11.9	10.6	10.0
		MCDBR	11.9	13.5	15.2	17.5	17.1	17.5	19.0	19.4	19.4	16.2	13.0	11.7
	5% WB	MCWBR	7.2	7.9	8.2	8.7	7.7	7.2	6.6	7.0	8.0	7.7	7.2	7.2
		MCWBR	11.3	13.1	14.5	16.7	15.6	16.0	17.4	17.5	17.8	14.9	12.7	11.1
(40) Clear-Sky Solar Irradiance	taub		0.251	0.259	0.275	0.321	0.331	0.332	0.335	0.355	0.316	0.286	0.258	0.245
		taud	2.505	2.516	2.537	2.404	2.424	2.464	2.470	2.374	2.459	2.545	2.533	2.494
	Ebn at Noon		865	931	961	933	932	929	921	887	898	882	847	827
		Edh at Noon	60	74	86	108	111	107	105	111	92	72	58	54
(44) All-Sky Solar Radiation	RadAvg		1.45	2.43	3.62	4.84	5.72	6.60	7.22	5.96	4.29	2.72	1.61	1.16
	RadStd		0.09	0.17	0.29	0.27	0.51	0.44	0.37	0.33	0.33	0.24	0.13	0.08

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page